

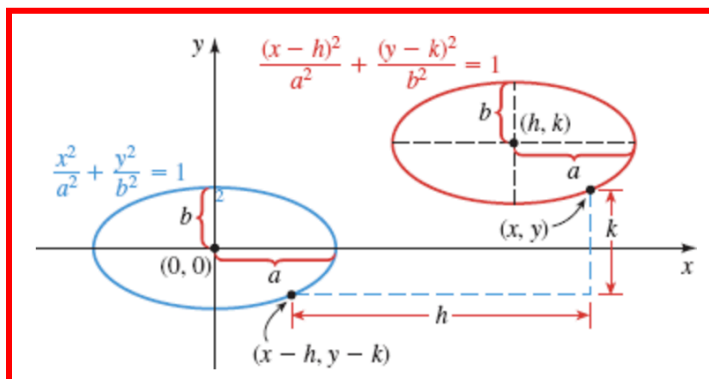
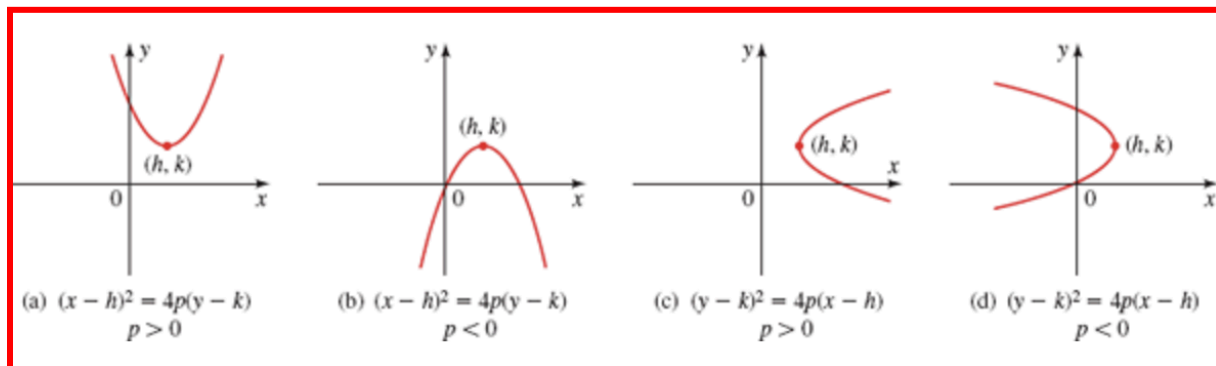
Shifting Graphs of Equations

If h and k are positive real numbers, then replacing x by $x - h$ or by $x + h$ and replacing y by $y - k$ or by $y + k$ has the following effect(s) on the graph of any equation in x and y .

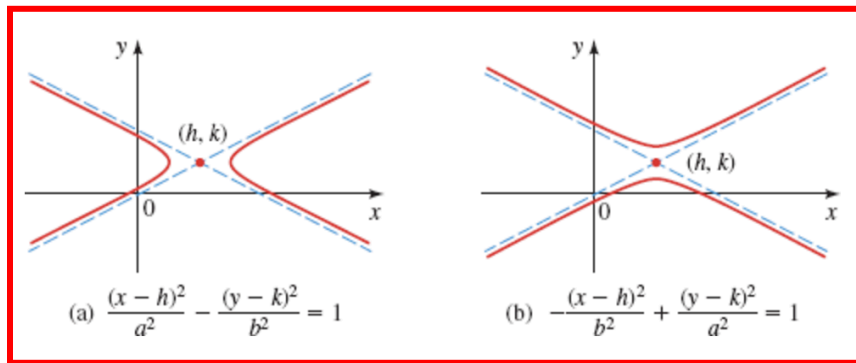
Replacement	How the graph is shifted
1. x replaced by $x - h$	Right h units
2. x replaced by $x + h$	Left h units
3. y replaced by $y - k$	Upward k units
4. y replaced by $y + k$	Downward k units

Shifted Ellipses

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

**Shifted Parabolas**

Shifted Hyperbolas



General Equation of a Shifted Conic

The graph of the equation $Ax^2 + Cy^2 + Dx + Ey + F = 0$, where A and C are not both 0, is a **conic** or a **degenerate conic**. In the **non degenerate cases** the graph is

1. a **parabola** if A or C is 0,
2. an **ellipse** if A and C have the **same sign** (or a **circle** if $A = C$),
3. a **hyperbola** if A and C have **opposite signs**.

Example 1: An equation of an ellipse is given.

- (a) Find the center, vertices, and foci of the ellipse.
- (b) Determine the lengths of the major and minor axes.
- (c) Sketch a graph of the ellipse.

i. $\frac{x^2}{4} + (y + 2)^2 = 1$

ii. $9x^2 - 54x + y^2 + 2y + 46 = 0$

Example 2: An equation of a parabola is given.

(a) Find the vertex, focus, and directrix of the parabola.

(b) Sketch a graph showing the parabola and its directrix.

i. $(y + 1)^2 = 16(x - 3)$

ii. $x^2 - 6x - 12y + 33 = 0$

Example 3: An equation of a hyperbola is given.

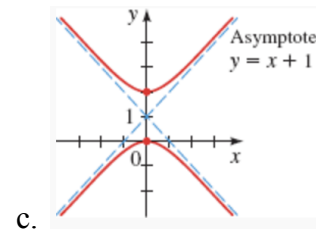
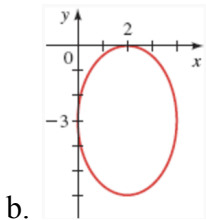
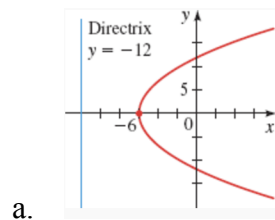
(a) Find the center, vertices, foci, and asymptotes of the hyperbola.

(b) Sketch a graph showing the hyperbola and its asymptotes.

i. $\frac{(y-1)^2}{25} - (x + 3)^2 = 1$

ii. $36y^2 + 72y - 4x^2 + 32x + 116 = 0$

Example 4: Find an equation for the conic whose graph is shown.



Example 5: Find the standard equation for the conic section with the given properties.

a. The ellipse with center $(2, -3)$, vertices $(-8, -3)$ and $(12, -3)$, and foci $(-4, -3)$ and $(8, -3)$.

b. The hyperbola with vertices $(-1, -1)$ and $(5, -1)$ and foci $(-4, -1)$ and $(8, -1)$.

c. The parabola with focus $(1, 3)$ and directrix $x = 3$

d. The hyperbola with foci $(1,-5)$ and $(1,5)$ that passes through the point $(1,4)$.

e. The parabola that passes through the point $(6,1)$, with vertex $(-1,2)$ and horizontal axis of symmetry

f. The ellipse with foci $(3,-4)$ and $(3,4)$, and x -intercepts 0 and 6